

Data Collection and Preprocessing Phase

Field	Detail
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Team ID	Individual
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues such as missing values and category imbalance, and implements resolution plans to ensure accurate and reliable analysis of global AI adoption trends, policy impact, and equity gaps in education across 1,360 records and 16 attributes.

Section	Description
Data Overview	The dataset contains 1,360 global AI-in-education records with 16 columns: year, month, country, region, student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, top_ai_tool, internet_penetration_pct, education_index, government_ai_policy, urban_ai_usage_pct, rural_ai_usage_pct, ai_in_curriculum, and gender_gap_ai_usage_pct. It covers 10 countries across 6 regions (Asia, North America, Europe, South America, Africa, Oceania). Records span monthly observations from 2015 to 2026. Student AI usage ranges from 0% to 65% (avg 30.99%), teacher AI usage from 0% to 60% (avg 29.91%), and school-level AI adoption from 0% to 62% (avg 29.41%). 15 of 16 columns are fully populated; government_ai_policy contains 441 missing entries (32.4%).
Data Cleaning	No duplicate records found — all 1,360 country-year-month combinations are unique. Missing values are confined to government_ai_policy (441 of 1,360 rows, 32.4%), which represent countries/periods with no formal AI-in-education policy rather than a data-collection error; these are resolved by mapping nulls to an explicit “None” category alongside the existing “Draft” and “Active” statuses. Categorical columns (country, region, top_ai_tool, ai_in_curriculum) contain clean, consistent labels with no mixed-case or spelling variants. Numerical columns (student/teacher/school AI usage %, internet penetration %, urban/rural usage %, gender gap %, education_index) fall within expected bounds. Dataset is confirmed

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	clean and Tableau-ready after null handling.
Data Transformation	Filter records by country, region, year, and government_ai_policy for focused segment analysis. Sort by student_ai_usage_pct descending to identify leading adoption markets, and by schools_ai_adoption_pct to surface leading school systems. Pivot data to create country-wise and region-wise year-over-year aggregations of average AI usage and adoption. Create calculated columns: Urban-Rural Usage Gap = (urban_ai_usage_pct – rural_ai_usage_pct); Student-Teacher Adoption Ratio = (student_ai_usage_pct / teacher_ai_usage_pct); Adoption Tier (Low: 0–25, Mid: 26–45, High: 46–65) based on student_ai_usage_pct; Policy Status (Clean) = IFNULL([government_ai_policy], “None”).
Data Type Conversion	year and month confirmed as int64 and used to build time-series axes. student_ai_usage_pct, teacher_ai_usage_pct, schools_ai_adoption_pct, avg_daily_ai_usage_min, internet_penetration_pct, urban_ai_usage_pct, rural_ai_usage_pct, and gender_gap_ai_usage_pct (int64, whole-number percentages/minutes) are correctly typed for aggregation. education_index (float64, range 0.52–0.94) confirmed for precise averaging and correlation analysis. country, region, top_ai_tool, government_ai_policy, and ai_in_curriculum confirmed as str and loaded as Tableau Dimensions. No type mismatches detected.
Column Splitting and Merging	Merge year and month into a single Period field (formatted YYYY-MM) to support clean time-series axes across the 2015–2026 span. Merge country and region into a Location Label (e.g. “India – Asia”) to support geographic drill-downs. Convert ai_in_curriculum (Yes/No) into a Boolean Curriculum Integrated flag for rate calculations. Combine government_ai_policy with ai_in_curriculum into a Policy-Curriculum Alignment tag (e.g. “Active – Yes”) to explore how policy adoption translates into classroom practice.
Data Modeling	Primary table: Education AI Adoption (1,360 rows, 16 columns) used as single source of truth. Tableau calculated measures: Avg Student Usage = AVG(student_ai_usage_pct) = 30.99%; Avg Teacher Usage = 29.91%; Avg School Adoption = 29.41%; Avg Internet Penetration = 74.70%; Avg Education Index = 0.776; Avg Gender Gap = 5.41 points; Active Policy Rate ≈ 34.85% (474/1,360); Draft Policy Rate ≈ 32.72% (445/1,360); No-Policy Rate ≈ 32.43% (441/1,360); Curriculum Integration Rate ≈ 50.51% (687/1,360). Country distribution is evenly balanced at 136 records each across all 10 countries. Region distribution is skewed toward Asia (408 records, 30%), which spans three countries (China, India, Pakistan) versus one to two countries for other regions.
Save Processed Data	Original dataset (Global_AI_in_Education.csv, 1,360 rows x 16 columns) saved as primary source — no rows removed, as the dataset required only null-handling on one column. Processed CSV with

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	added calculated columns (Urban-Rural Usage Gap, Student-Teacher Adoption Ratio, Adoption Tier, Period, Location Label, Policy-Curriculum Alignment, Policy Status Clean) exported as ai_education_processed.csv for Tableau ingestion. Tableau workbook saved as global_ai_education_analysis.twbx with embedded data source. All files version-controlled in the GitHub repository with preprocessing steps documented in README.md.